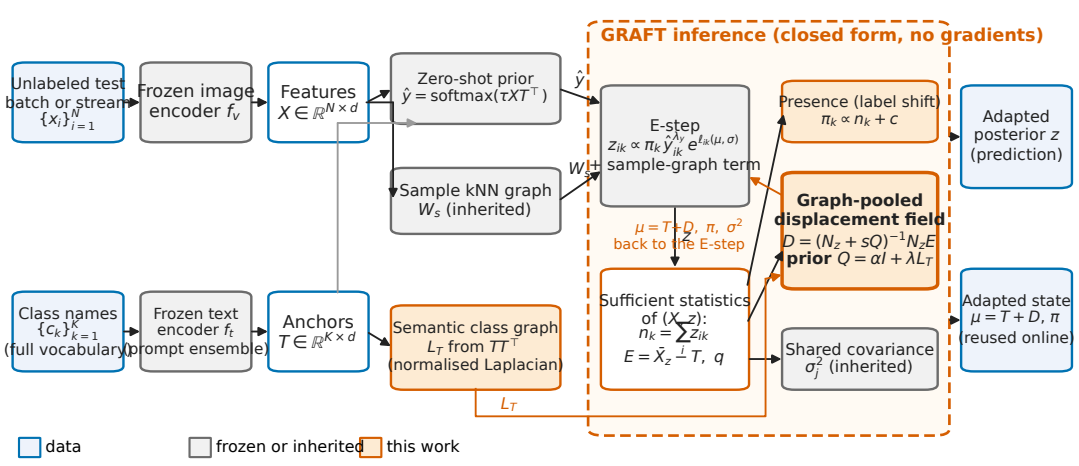
GRAFT inference (closed form, no gradients)
Unlabeled test batch or stream $\{x_i\}_{i=1}^{N}$
Frozen image encoder f_v
Features $X \in \mathbb{R}^{N \times d}$
Zero-shot prior $\hat{y} = \mathrm{softmax}(\tau X T^\top)$
$\hat{y}$
Sample kNN graph W_s (inherited)
W_s
E-step $z_{ik} \propto \pi_k \hat{y}_{ik}^{\lambda_y} e^{\ell_{ik}(\mu,\sigma)}$ + sample-graph term
$\mu = T + D,\ \pi,\ \sigma^2$ back to the E-step
Presence (label shift) $\pi_k \propto n_k + c$
Graph-pooled displacement field $D = (N_z + sQ)^{-1} N_z E$ prior $Q = \alpha I + \lambda L_T$
Adapted posterior z (prediction)
Sufficient statistics of (X, z): $n_k = \sum z_{ik}$, $E = \bar{X}_z - T,\ q$
Shared covariance σ_j^2 (inherited)
Adapted state $\mu = T + D,\ \pi$ (reused online)
Class names $\{c_k\}_{k=1}^{K}$ (full vocabulary)
Frozen text encoder f_t (prompt ensemble)
Anchors $T \in \mathbb{R}^{K \times d}$
Semantic class graph L_T from $TT^\top$ (normalised Laplacian)
L_T
data
frozen or inherited
this work